

Lab 1

01.2: ARP, Wireshark, Netsim

1. ARP (linux.cs.pdx.edu)

- Use the `ip address` command to find the IPv4 address and hardware address of the local ethernet card interface (Typically beginning with `eth`, `ens`, or `enp`).

Answer: **IPv4:** 131.252.208.103, **MAC address:** ether 52:54:00:13:a0:c6

- What is the default router's IP address (e.g. the gateway address for the default route 0.0.0.0/0)

Answer: 131.252.208.1

- What is the name of the default router and its hardware address?

Answer: Name: router.seas.pdx.edu, MAC address: 00:00:5e:00:01:01

- How many entries are there in the ARP table?

Answer: 33

2. -

- List any IP addresses that share the same hardware address

Answer:

131.252.208.81 and 131.252.208.171 share cc:aa:77:07:f2:7a

131.252.208.118 and 131.252.208.53 share 52:54:00:30:e3:f2

- How many less hardware addresses are there than IP addresses in the ARP table?

Answer: 2 fewer hardware addresses than IP addresses

Use a single command-line to create a file that contains each IP address that appears in the machine's ARP table and places the results in a file called `arp_entries`. The command should be similar to the one below:

```
arp -an | awk -F '...' '{print ...}' > arp_entries
```

- Include the command in your lab notebook:

Answer: `arp -an | awk -F '[]' '{print $2}' > arp_entries`

Examine the `arp_entries` file:

- What network prefix do most of the IP addresses in the ARP table share?

Answer: 131.252.208

3. ARP (Cloud)

- Find the IP address and hardware address of the local ethernet card interface (Typically beginning with `eth`, `ens`, or `enp`).

Answer: ip: 10.138.0.2, mac address 42:01:0a:8a:00:02

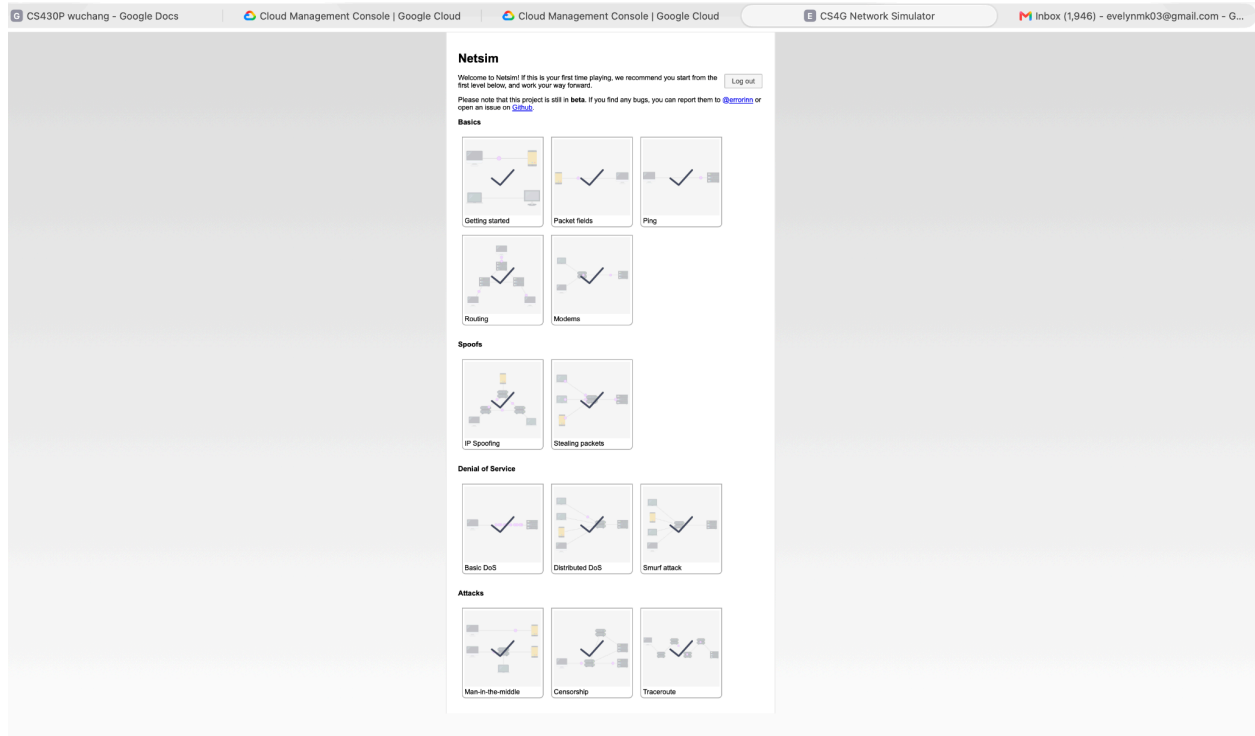
- What is the default router's IP address (e.g. the gateway address for the default route 0.0.0.0/0)

Answer: 10.138.0.1

- What is the default router's hardware address?

Answer: 42:01:0a:8a:00:01

4. Netsim



01.3: Cloud networking

3. Scan targets for services

- Show a screenshot of the output for the scan for your lab notebook.

```
evelynnng@course-vm:~$ nmap 10.138.0.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-05 18:09 UTC
Nmap scan report for course-vm.c.cloud-nguyen-evelynnng.internal (10.138.0.2)
Host is up (0.00071s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
3389/tcp  open  ms-wbt-server

Nmap scan report for instance-20261005-175557.c.cloud-nguyen-evelynnng.internal (10.138.0.3)
Host is up (0.00033s latency).
Not shown: 990 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
631/tcp   open  ipp
3306/tcp  open  mysql

Nmap scan report for instance-20261005-175653.c.cloud-nguyen-evelynnng.internal (10.138.0.4)
Host is up (0.00039s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap scan report for instance-20261005-175914.c.cloud-nguyen-evelynnng.internal (10.138.0.5)
Host is up (0.00079s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (4 hosts up) scanned in 4.23 seconds
evelynnng@course-vm:~$
```

5. Navigating default networks

- How many subnetworks are created initially on the `default` network? How many regions does this correspond to? (Use a pipe to pass output to `grep` in order to return specific lines of output and then another to pass output to `wc` to count them: `| grep default | wc -l`)

Answer: 84 subnetworks, corresponding to 84 regions

- Given the CIDR prefix associated with each subnetwork, how many hosts does each subnetwork support?

Answer: The CIDR is 20. So leaving 12 bits for host -> $2^{12} = 4096$ addresses

- Which CIDR subnetworks are these instances brought up in? Do they correspond to the appropriate region based on the prior commands?

Answer: instance-1 belongs to us-west1 10.138.0.0/20 , instance-2 belongs to us-central1 10.128.0.0/20. They both respond to appropriate regions.

- From `instance-1`, perform a ping to the Internal IP address of `instance-2`. Take a screenshot of the output.

```
evelynnng@instance-1:~$ ping 10.128.0.2
PING 10.128.0.2 (10.128.0.2) 56(84) bytes of data.
64 bytes from 10.128.0.2: icmp_seq=1 ttl=64 time=35.9 ms
64 bytes from 10.128.0.2: icmp_seq=2 ttl=64 time=35.0 ms
64 bytes from 10.128.0.2: icmp_seq=3 ttl=64 time=35.0 ms
64 bytes from 10.128.0.2: icmp_seq=4 ttl=64 time=34.9 ms
64 bytes from 10.128.0.2: icmp_seq=5 ttl=64 time=35.1 ms
64 bytes from 10.128.0.2: icmp_seq=6 ttl=64 time=34.9 ms
64 bytes from 10.128.0.2: icmp_seq=7 ttl=64 time=31.6 ms
64 bytes from 10.128.0.2: icmp_seq=8 ttl=64 time=31.7 ms
64 bytes from 10.128.0.2: icmp_seq=9 ttl=64 time=31.7 ms
64 bytes from 10.128.0.2: icmp_seq=10 ttl=64 time=31.5 ms
64 bytes from 10.128.0.2: icmp_seq=11 ttl=64 time=31.5 ms
64 bytes from 10.128.0.2: icmp_seq=12 ttl=64 time=31.7 ms
64 bytes from 10.128.0.2: icmp_seq=13 ttl=64 time=31.5 ms
64 bytes from 10.128.0.2: icmp_seq=14 ttl=64 time=31.5 ms
64 bytes from 10.128.0.2: icmp_seq=15 ttl=64 time=31.6 ms
64 bytes from 10.128.0.2: icmp_seq=16 ttl=64 time=31.6 ms
64 bytes from 10.128.0.2: icmp_seq=17 ttl=64 time=31.7 ms
64 bytes from 10.128.0.2: icmp_seq=18 ttl=64 time=31.9 ms
64 bytes from 10.128.0.2: icmp_seq=19 ttl=64 time=31.6 ms
```

- From the figure in the previous step. What facilitates this connectivity: the virtual switch or the VPN Gateway?

Answer: virtual switch because 2 instances are both inside google cloud VPC

6. Creating custom networks

- Take a screenshot of the new subnets created in `custom-network1` alongside the default subnetworks in those regions assigned to the `default` network.

```
evelynng@cloudshell:~ (cloud-nguyen-evelynng)$ gcloud compute networks subnets list --regions=us-central1,europe-west1
NAME: default
REGION: europe-west1
NETWORK: default
RANGE: 10.132.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: subnet-europe-west-192
REGION: europe-west1
NETWORK: custom-network1
RANGE: 192.168.5.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: default
REGION: us-central1
NETWORK: default
RANGE: 10.128.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: subnet-us-central-192
REGION: us-central1
NETWORK: custom-network1
RANGE: 192.168.1.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:
```

- Explain why the result of this ping is different from when you performed the ping to [instance-2](#).

Answer: the ping fails because the instances are on separate VPC networks. Instance-1 is on the default network, while instances 3 and 4 are on custom-network1. Separate VPC networks do not have connectivity between each other by default.

- Take screenshots of all 4 instances in the UI including the network they belong to.

☐	☒	instance-1	us-west1-c	10.138.0.6 (nic0)	8.229.126.46 (nic0)	default	SSH	▼	⋮
☐	☒	instance-2	us-central1-a	10.128.0.2 (nic0)	136.111.154.186 (nic0)	default	SSH	▼	⋮
☐	☒	instance-3	us-central1-a	192.168.1.2 (nic0)	34.135.44.74 (nic0)	custom-network1	SSH	▼	⋮
☐	☒	instance-4	europe-west1-d	192.168.5.2 (nic0)	34.156.9.8 (nic0)	custom-network1	SSH	▼	⋮

- Take a screenshot of the subnetworks created for the `custom-network1` network and some of the subnetworks of the `default` network showing their regions, internal IP ranges and Gateways

```
AME: default
EGION: europe-west1
ETWORK: default
ANGE: 10.132.0.0/20
ATEWAY_ADDRESS: 10.132.0.1

AME: subnet-europe-west-192
EGION: europe-west1
ETWORK: custom-network1
ANGE: 192.168.5.0/24
ATEWAY_ADDRESS: 192.168.5.1

AME: default
EGION: us-central1
ETWORK: default
ANGE: 10.128.0.0/20
ATEWAY_ADDRESS: 10.128.0.1

AME: subnet-us-central-192
EGION: us-central1
ETWORK: custom-network1
ANGE: 192.168.1.0/24
ATEWAY_ADDRESS: 192.168.1.1
```

```
NAME: default
REGION: us-central1
NETWORK: default
RANGE: 10.128.0.0/20
GATEWAY_ADDRESS: 10.128.0.1

NAME: subnet-us-central-192
REGION: us-central1
NETWORK: custom-network1
RANGE: 192.168.1.0/24
GATEWAY_ADDRESS: 192.168.1.1
evelynnng@cloudshell:~ (cloud-nguyen-evelynnng) $ gcloud compute networks subnet
--regions=us-central1,europe-west1
NAME: default
REGION: europe-west1
NETWORK: default
RANGE: 10.132.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: subnet-europe-west-192
REGION: europe-west1
NETWORK: custom-network1
RANGE: 192.168.5.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:

NAME: default
REGION: us-central1
NETWORK: default
RANGE: 10.128.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
UTILIZATION_DETAILS:
```

